

A Segment-Aware Transfer Learning Framework for Reliable Early-Stage Cost Estimation of Building Retrofit Projects

Nahyun Kwon ^a; Sojin Park ^b, Yonghan Ahn ^a, Ho Anh Thu Nguyen ^{b,*}

^a Department of Architectural Engineering, Hanyang University-ERICA, Ansan 15588, Republic of Korea

^b Department of Smart City Engineering, Hanyang University-ERICA, Ansan 15588, Republic of Korea

* Correspondence: thunguyen98@hanyang.ac.kr

Abstract

Reliable early-stage cost estimation is essential for building retrofit planning, yet prediction remains challenging because retrofit costs are often highly skewed and high-cost projects are underrepresented in available datasets. Conventional single-stage machine learning models may achieve acceptable overall accuracy but frequently underestimate extreme-cost projects, increasing the risk of budget shortfalls and unreliable investment decisions. To address this issue, this research proposes a segment-aware transfer learning framework for building retrofit cost estimation. The framework first classifies projects into low- and high-cost segments using a support vector machine classifier and then applies segment-specific artificial neural network regressors enhanced through transfer learning from data-rich to data-scarce cost segments. The framework was evaluated using a real-world dataset of 419 retrofit projects involving nursery schools and healthcare facilities, with direct construction costs ranging from approximately USD 5,000 to USD 800,000. Results show that the proposed framework achieved an average R^2 of 0.9566 and reduced MAPE and RMSE by 46% and 25%, respectively, compared with the best-performing single-stage baseline. The framework also reduced prediction errors for high-cost projects, mitigating the systematic underestimation observed in conventional models. SHAP analysis further revealed that insulation, window replacement, HVAC systems, and renewable energy measures were key cost-driving factors. The proposed framework provides a reliable and interpretable decision-support approach for early-stage retrofit cost planning under skewed and data-scarce conditions.

Keywords: Building retrofit; Cost estimation; Transfer learning; Segment-aware modeling; Decision support

1. Introduction

As global awareness of environmental degradation intensifies and natural resources become increasingly scarce, the construction and operation of buildings remains one of the primary contributors to global energy consumption and greenhouse gas emissions [1-3]. According to recent studies, the building

sector accounts for approximately 40% of worldwide energy usage, with the operational phase of buildings responsible for the majority of this demand [2, 4, 5]. This growing energy demand highlights the importance of interventions for improving energy efficiency, particularly within the context of existing building stocks, which often exhibit outdated systems and poor thermal performance. As a result, building retrofitting approaches, such as energy-efficient renovations, and system upgrades, have emerged as a cornerstone strategy for achieving effective energy consumption and sustainability of the deteriorated buildings [2, 6-8]. Beyond energy inefficiency, deteriorated buildings commonly experience structural deterioration and functional obsolescence [9-11], which further diminish occupant comfort and increase maintenance costs [2, 12]. Despite the clear environmental and economic benefits, the implementation rate of retrofitting projects remains relatively low. One critical barrier lies in the uncertainty associated with project costs. Therefore, accurate, reliable, and interpretable cost forecasting is essential for stakeholders to effectively evaluate retrofitting feasibility, secure investment, and align technical decisions with broader financial strategies [12].

The current cost estimation for retrofitting projects often falls short in accounting for this necessity. Traditional approaches rely heavily on historical averages, cost ratios, or expert heuristics, all of which may introduce significant biases and fail to capture the complex interplay of variables that influence construction costs. In response, a number of studies have explored data-driven alternatives, including Case-Based Reasoning (CBR) [13], Genetic Algorithms (GA) [2], and fuzzy-based regression model [14], which attempt to optimize prediction by learning and identifying key feature relationships. While these models have improved upon basic estimation techniques, they are often constrained by their dependence on specific feature structures, limited scalability, and difficulties in handling non-linear relationships and distributional extremes.

To address these limitations, recent studies have employed advanced ML techniques such as boosting algorithms [15] and deep neural networks [16-18], which enhance prediction accuracy by capturing complex non-linear relationships between input variables and construction costs. Nevertheless, these ML models face two key challenges. First, they are often trained on datasets with limited sample sizes [15] or incomplete project details (i.e., limited independent variables) [16, 18]; in some cases, studies rely solely on generative or simulated data [17], lacking validation from large-scale field data. These limitations increase the risk of overfitting or underfitting. Second, the use of a single ML model struggles to provide accurate predictions for various projects with a wide cost range [19, 20]. While acceptable performance is often achieved for cost ranges that are well represented in the training data, predictions for exceptionally high-cost projects remain inaccurate [19, 20]. This shortcoming can lead to systematic underestimation and, consequently, underfunding

of retrofitting projects. Therefore, there is a clear need for modeling strategies that can robustly handle extreme cost distributions and maintain predictive reliability under data-scarce conditions.

To address these limitations, this research proposes a segment-aware transfer learning framework for early-stage building retrofit cost estimation across a wide cost spectrum. The framework aims to improve predictive reliability for high-cost projects by first classifying projects into cost segments and then transferring knowledge from data-rich low-cost projects to data-scarce high-cost projects. Specifically, the objectives of this research are to: (i) develop a two-stage framework that integrates SVM-based cost range classification with ANN-based cost estimation; (ii) systematically evaluate multiple transfer learning strategies and cost cutoffs under data-scarce and skewed cost conditions; and (iii) interpret segment-specific cost drivers using SHAP analysis to support transparent retrofit cost decision-making. The key contributions and novelties of this paper can be summarized as follows:

- A segment-aware transfer learning framework for high-cost retrofit estimation: This research proposes a two-stage framework that combines cost range classification with transfer learning-enhanced regression for building retrofit cost estimation. By transferring knowledge from abundant low-cost projects to data-scarce high-cost segments, the framework improves predictive reliability and reduces underestimation risk for high-cost retrofit projects. The effects of different cost cutoffs and transfer learning strategies are also systematically evaluated to enhance robustness and practical applicability.
- Explainable ML to support transparent and expert-informed decision-making: SHAP-based interpretability is incorporated to quantify the influence of major cost-driving factors across different cost segments. This improves model transparency and helps practitioners understand how retrofit, design, and energy-related variables affect cost predictions, thereby supporting informed decision-making during early project planning.
- A comprehensive real-world retrofitting cost dataset for practical decision support: The research utilizes a dataset of 419 real-world retrofitting projects containing detailed building characteristics and retrofit work quantities. Covering a broad cost range, the dataset supports robust model development and enhances the practical relevance of the proposed framework for real-world retrofitting cost estimation.

2. Related studies

2.1 Construction cost prediction

With the rapid advancement of data-driven methodologies, an increasing number of studies have utilized historical datasets to develop construction cost prediction models using various estimation techniques. Table 1 summarizes representative studies by analyzing their dataset characteristics, prediction algorithms, and reported limitations.

Traditional approaches such as multiple linear regression (MLR), GA, and CBR have been widely applied and have demonstrated reasonable predictive performance. For example, Kwon et al. [12] integrated GA with CBR to estimate the maintenance costs of residential buildings, achieving a mean absolute error of 18% on the validation dataset. Nevertheless, the authors noted that the limited size of historical data could introduce significant uncertainty when predicting cases that differ substantially from previously observed projects.

Methods such as MLR, GA, and CBR have been widely applied and have demonstrated reasonable predictive accuracy. For example, Kwon et al. [12] integrated GA with CBR to estimate the maintenance costs of residential buildings, achieving a mean absolute error of 18% on the validated dataset. However, the authors also acknowledged that the limited availability of historical data may lead to substantial uncertainty when predicting cases that differ from previously observed projects. Similar dataset-related limitations were reported by Mahmoud et al. [21], who employed MLR to predict the maintenance costs of aging heritage buildings. Their model was developed using only 33 data samples with three input variables. Although satisfactory performance was observed on the test set, the restricted number of variables and small dataset limit the model's reliability and practical applicability. Furthermore, the inherent limitations of linear regression-based approaches were highlighted by Li and Guo [22], whose findings demonstrated that ML algorithms generally achieve superior predictive performance compared with traditional methods due to their ability to capture complex non-linear relationships among variables.

Recently, owing to their superior predictive performance and the increasing availability of data, ML algorithms (e.g., ANN, LSTM, and SVM) have been increasingly adopted for construction cost prediction. As summarized in Table 1, studies employing ML generally utilize larger datasets and wider data ranges than those relying on traditional methods [23-27], and the number of input variables is no longer severely constrained. Nevertheless, with the exception of the study by Kim et al. [24], which utilized 3,444 data samples, most existing studies still rely on fewer than 200 data points, a scale that remains insufficient for fully exploiting the learning capacity of ML models [28]. In terms of algorithm selection, ANN and SVM are the most frequently employed approaches.

However, as illustrated in the results reported by Wu et al. [26] and Al-Gahtani et al. [27], while these models achieve satisfactory accuracy for projects with normal cost levels, prediction errors increase substantially for extreme values at both ends of the cost spectrum. Similar limitations of ML models under data-scarce conditions have been observed in other prediction tasks characterized by wide output ranges [29-31].

Among existing studies on construction cost prediction, only a limited number specifically address retrofitting projects [12, 32-34]. Consistent with the broader construction cost literature, these studies commonly exhibit a noticeable degradation in predictive accuracy for high-cost projects compared with the distributions represented in their training datasets. Inaccurate estimation of high-cost projects, in particular, may lead to systematic underestimation, budget shortfalls, and subsequent cost overruns. Despite the practical importance of this issue, effective methodological solutions for addressing prediction uncertainty in extreme cost ranges remain lacking due to the scarcity of data on high-cost projects.

Table 1. Recent studies in construction cost prediction

Ref	Objectives	Data	Range	Method	Input	Disadvantages
[12]	Maintenance cost of aging residential buildings	90	3,150.6-5,254.5 USD	CBR, GA	Building coverage ratio, floor area ratio, number of buildings, number of floors, number of households, parking lots, maintenance area, completion year	Strong dependence on historical data
[21]	Maintenance cost of aging heritage buildings	33	10,000-70,000 MYR	MLR	Age of buildings, gross floor area, building performance index	Limited dataset size and a small number of variables
[22]	Maintenance cost of aging university buildings	-	-	LR, MLR, ANN	Building age, number of floors, number of classrooms, elevators	Small number of study objects; strong dependence on historical data
[23]	Maintenance cost of buildings	100	2,500-14,500 USD	ANN	System type, system age, maintenance frequency, building type, environmental conditions	High implementation complexity; limited cost range
[25]	Green building construction cost	148	1,400-3,400 Yuan	PCA-SVM	Project characteristics, green building attributes, market and location factors	The model mainly focuses on environmental factors, while neglecting material quality and the professional competence
[26]	Green building construction cost	126	1,400-3,400 Yuan	LSSVM	Project characteristics, green building attributes, market and location factors	Large prediction errors for projects with costs exceeding 2,800 Yuan

[27]	Final construction cost	135	0-30,000,000 KRW	ANN	Area-related variables, parking lots, building age	Large prediction errors for projects with costs exceeding 25,000,000 KRW
------	-------------------------	-----	------------------	-----	--	--

Notes: CBR: Case-Based Reasoning; GA: Genetic Algorithm; MLR: Multiple Linear Regression; LR: Linear Regression; ANN: Artificial Neural Network; LSTM: Long Short-Term Memory; PCA: Principal Component Analysis;

2.2 TL in construction

Recently, TL has emerged as an effective approach for transferring knowledge across domains to specialize models for specific prediction tasks. In the construction field, TL frameworks have been widely adopted to improve data efficiency by leveraging information from large source datasets. Typically, a source model is first trained on a well-represented source domain, and the learned knowledge is subsequently adapted or fine-tuned for a target domain characterized by limited data availability [30].

In the context of energy prediction, Gao et al. [35] utilized high-quality building datasets to transfer knowledge for predicting energy consumption in buildings with scarce data, demonstrating that TL frameworks based on LSTM and CNN outperformed traditional ML approaches. In concrete engineering, Ford et al. [36] employed a large dataset of natural aggregate concrete to predict the compressive strength of recycled aggregate concrete using TL. Similar TL approaches have also been applied to predict the lateral strength of reinforced concrete columns [37], steel corrosion in concrete [38], and the compressive strength of ultra-high-performance geopolymer [30]. Beyond regression tasks, TL has also been extensively applied in construction-related image processing applications, including crack detection in structures [39], safety monitoring [40], and risk assessment for tunnel construction [41]. Collectively, these studies demonstrate that TL can significantly improve predictive performance in data-scarce scenarios, highlighting its potential for enhancing cost prediction in high-cost retrofitting projects.

3. Methodology

To develop and systematically evaluate the TL-enhanced two-stage framework, this research follows three principal stages: data preparation, model development, and model evaluation and interpretation. The overall research workflow is illustrated in Figure 1.

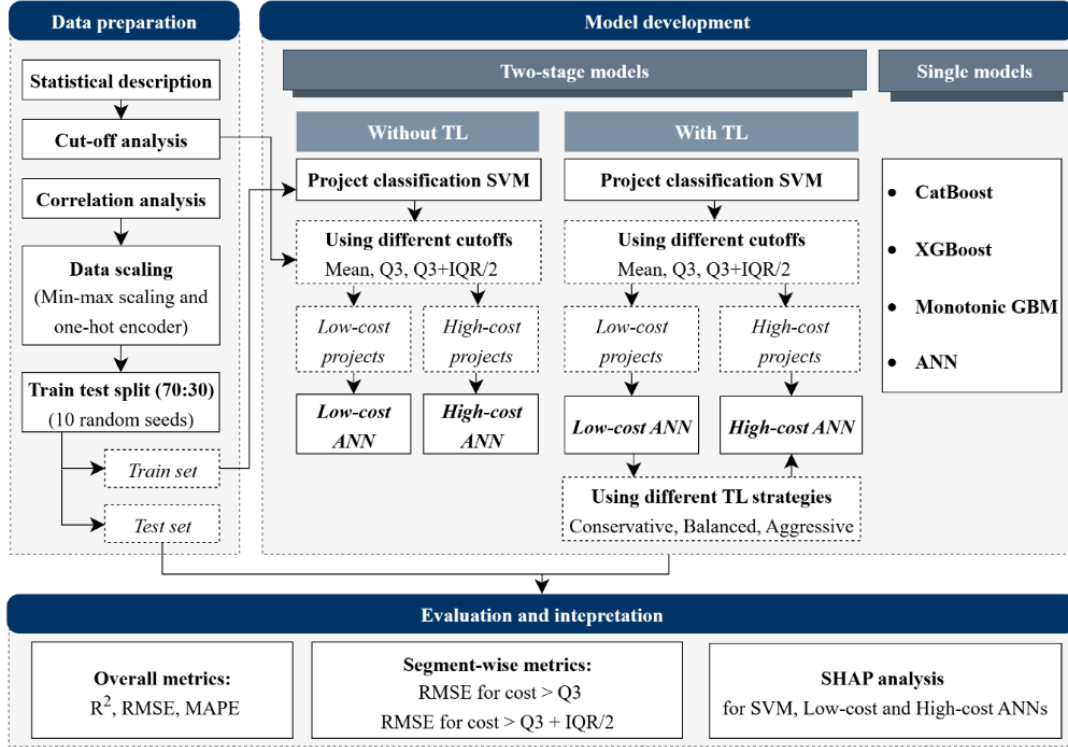


Figure 1. Research workflow

First, the dataset was prepared and subjected to statistical analysis, including an examination of data distributions and correlations between input and output features. This exploratory analysis informs the selection of appropriate data preprocessing techniques and ML algorithms and further supports the identification of suitable cut-off points for developing the two-stage models.

Second, to ensure a comprehensive evaluation of model performance, multiple state-of-the-art algorithms were implemented, including single-stage models (i.e., CatBoost, XGBoost, monotonic GBM, and ANN) and a set of two-stage models that integrated SVM-based classification with ANN-based regression, both with and without TL.

Finally, performance of the models was evaluated using R-square score (R^2), (mean absolute percentage error) MAPE, and root mean squared error (RMSE) computed over the entire test set and segment-wise RMSE to further investigate the predictive performance in high-cost range. The best-performing model was subsequently interpreted using SHAP analysis to provide insights into feature contributions and model behavior.

3.1 Data preparation

The dataset consists of 419 retrofitting projects involving nursery schools and health centers, with retrofitting construction costs spanning approximately USD 5,000 to USD 800,000. It comprises 21 inputs, including one categorical feature and 20 numerical features, and an output feature representing the retrofitting construction cost (Table 2). The input features capture key project attributes, such as building characteristics (e.g., roof area and floor area) and the quantity of energy-related components (e.g., cool-proof elements and HVAC systems).

Table 2. Statistical description of the dataset

Name	Abb.	Unit	Min	Max	Mean	Std dev	Skewness
Building type (nursery school and health center)	C1	-	-	-	-	-	-
Year of construction completion	N1	year	1966	2015	2,004.06	7.31	-1.70
Construction area	N2	m ²	35.68	3,638.74	247.88	215.46	9.95
Gross floor area	N3	m ²	35.68	2,997.24	387.56	379.63	3.73
Number of above-ground floors	N4	count	1.00	5.00	1.75	0.71	0.83
Number of underground floors	N5	count	0.00	2.00	0.19	0.41	1.79
External wall insulation	N6	m ²	0.00	2,100.00	266.54	261.04	1.38
Internal wall insulation	N7	m ²	0.00	725.00	39.37	104.05	3.25
Roof insulation work quantity	N8	m ²	0.00	750.00	131.87	144.56	1.08
Floor work quantity	N9	m ²	0.00	1,000.00	30.21	96.19	4.76
Window replacement cost	N10	USD	0.00	299,347.20	24,736.65	28,972.42	5.10
Door installation cost	N11	USD	0.00	26,020.80	2,624.40	3,583.10	2.48
Cool roof work quantity	N12	m ²	0.00	788.48	53.89	110.66	2.62
Heat recovery ventilation system work quantity	N13	m ²	0.00	4,070.00	202.26	306.76	6.23
High-efficiency HVAC System installation cost	N14	USD	0.00	135,432.00	15,578.79	16,118.77	2.66
High-efficiency boiler installation cost	N15	USD	0.00	30,967.20	1,912.04	3,957.51	4.42
LED lighting replacement work quantity	N16	m ²	0.00	2,800.00	132.63	245.91	5.46
Remote energy metering device installation quantity	N17	m ²	0.00	1,380.00	38.05	135.85	4.77
Renewable energy (solar PV power generation system) quantity	N18	m ²	0.00	260.00	14.62	34.22	3.34

Solar radiation control device	N19	m ²	0.00	190.00	1.80	11.95	10.97
Instantaneous water heater	N20	count	0.00	6.00	0.12	0.52	6.16
Direct construction cost (Output)	O	USD	5,065.87	795,941.29	148,271.19	104,937.87	2.23

Figure 2 presents the Pearson correlation matrix for the numerical input features (N1–N20) and the output variable (O), which represents the direct construction cost. Several features exhibit moderate to strong positive correlations with the output variable. Notably, window replacement cost (N10) shows the highest correlation with construction cost ($r=0.81$) followed by gross floor area (N3) ($r=0.66$), external wall insulation (N6) ($r=0.65$), high-efficiency HVAC system installation cost (N14) ($r=0.59$). These variables likely represent significant contributors to the total cost of energy-efficient retrofitting projects. Weak or near-zero correlations are observed for features such as solar radiation control device (N19) and instantaneous water heater (N20), suggesting a limited direct impact on the construction cost within this dataset.

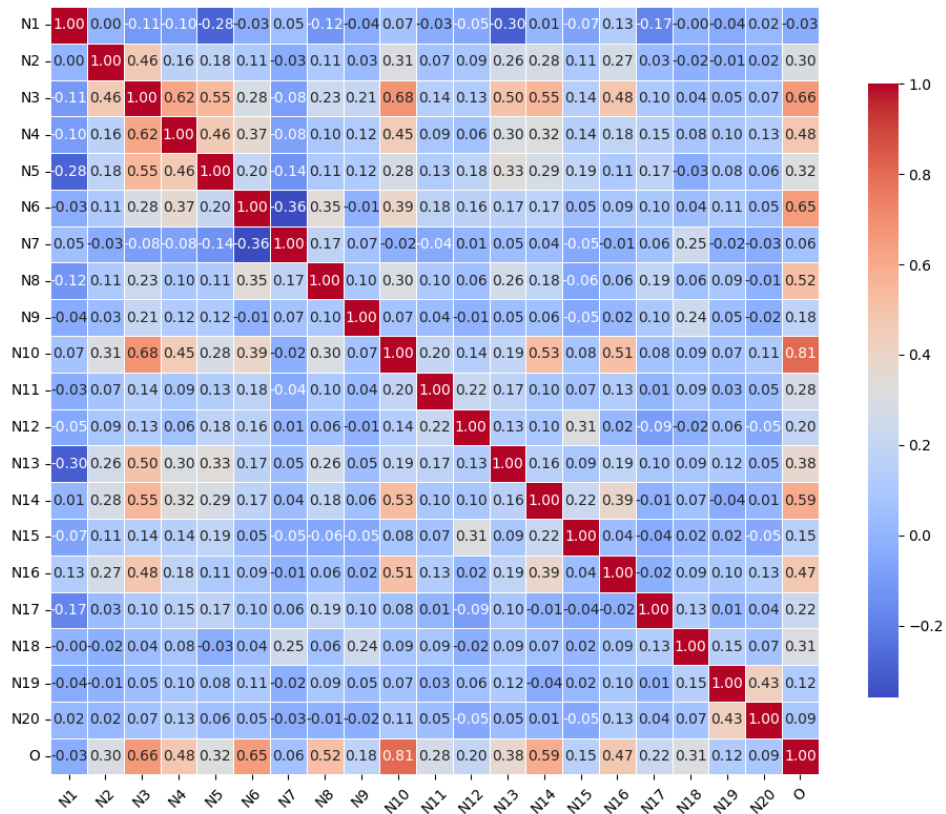


Figure 2. Correlation heatmap of the dataset

As illustrated in Figure 3, the distribution of retrofitting construction costs is strongly right-skewed, indicating the presence of a substantial number of extremely high-cost projects. In conventional single-stage modeling

approaches, an outlier removal step is often applied to improve model stability and predictive accuracy. However, excluding high-cost projects would substantially narrow the model's applicability, restricting it primarily to low- and medium-cost project segments. Moreover, in practical deployment, such models are prone to systematic underestimation of high-cost projects, thereby exposing decision-makers to an increased risk of cost overruns.

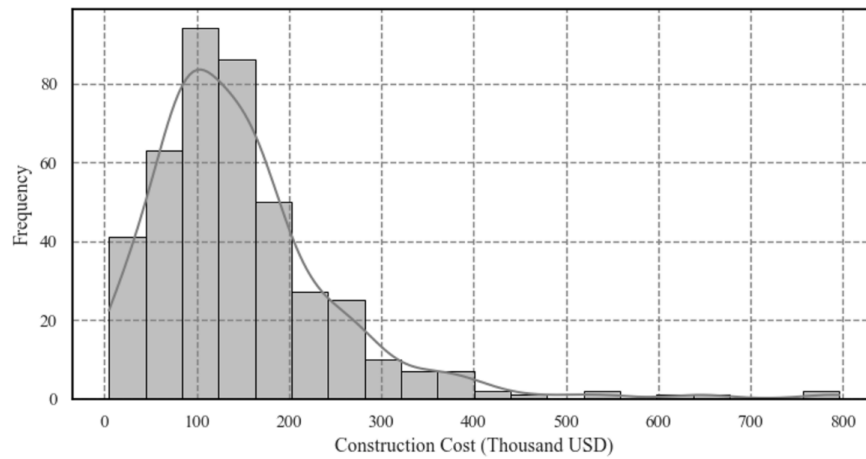


Figure 3. Data distribution of cost of retrofitting projects

Therefore, this research adopted a two-stage framework to partition projects into two cost segments, thereby enabling the use of segment-specific predictive models and avoiding the need to exclude extremely high-cost projects from the analysis. A critical step in developing an effective two-stage model is the identification of an appropriate cost cutoff. Projects with costs below a specified cut-off point are classified as low-cost, while those with costs equal to or greater than the cut-off point are categorized as high-cost. The selection of the cutoffs (including the mean, the third quartile (Q3), and an adjusted upper threshold ($Q3 + IQR/2$)), was motivated by two complementary considerations. First, the mean cost was adopted to prioritize class balance, thereby ensuring robust SVM training and stable classification performance [42]. Second, Q3 and $Q3 + IQR/2$ were introduced to investigate the effectiveness of TL when knowledge was transferred from a data-rich domain (low-cost segment) to a data-scarce domain (high-cost segment). The data segmentation for each cutoff strategy is summarized in Table 3.

Table 3. Description of different cutoffs

Cutoff	Value (USD)	Number of low-cost projects	Number of high-cost projects
Mean	148,271	243	176
Q3	180,676	314	105
Q3+IQR/2	228,626	352	67

For each random split, preprocessing parameters and model selection procedures were estimated using the training data only, while the testing subset was held out until final evaluation. Cost cutoffs and transfer learning strategies were then evaluated within this repeated hold-out setting to avoid information leakage from the test set.

3.2 Model development

The processed data was used for developing two categories of predictive models: single-stage models and two-stage models. The single-stage models comprise tree-based algorithms (i.e., CatBoost, XGBoost, and monotonic GBM) and neural network-based models (i.e., ANN).

To ensure a fair and consistent comparison across models, GridSearchCV was employed to systematically fine-tune the hyperparameters of all candidate models.

3.2.1 Single models

CatBoost

CatBoost is a GBDT-based algorithm designed for classification and regression tasks. It distinguishes itself from other GBDT methods through key innovations such as ordered boosting and efficient handling of categorical features [43]. Traditional GBDTs suffer from prediction shift due to using the same data for gradient estimation and model building, leading to target leakage [44]. CatBoost addresses this with an ordered boosting scheme that reduces gradient bias and overfitting. For categorical features, it replaces conventional preprocessing (e.g., one-hot encoding) with a modified target statistics approach that leverages random permutations and averaging across iterations [45]. Additionally, CatBoost employs oblivious trees (symmetric, balanced trees with uniform splitting rules) which enhance inference speed and reduce overfitting risk [46].

XGBoost

With changes to the loss function and loss optimization procedure, XGBoost [47] outperforms traditional gradient boosting algorithms. XGBoost is versatile, supporting classification, with regression, ranking, and user-defined functions, as well as distributed training across multiple machines. It is dedicated to pushing boosting trees past their computational limit, allowing engineers to achieve the engineering goals of faster computation and excellent performance.

Monotonic GBM

To establish a robust and domain-consistent baseline for wide-range construction cost prediction, a monotonic GBM was implemented. The model incorporates prior engineering knowledge by enforcing monotonic relationships between selected size- and quantity-related variables and project cost, thereby improving stability and interpretability.

Let $x = (x_1, x_2, \dots, x_d)$ denote the input feature vector and y the construction cost. The GBM prediction model is expressed as:

$$\hat{y} = f(x) = \sum_{m=1}^M \gamma_m h_m(x) \quad (1)$$

Where $h_m(\cdot)$ represents a regression tree at boosting iteration m , and γ_m is its corresponding weight.

Monotonicity constraints were applied to selected numerical features $S = \{N3, N4, N6, N8, N10, N16\}$, representing project scale and quality indicators, such that:

$$\frac{\partial f(x)}{\partial x_j} \geq 0 \quad (2)$$

This ensures that predicted costs do not decrease as these variables increase, while unconstrained features retain modeling flexibility.

ANN

ANN is a nonlinear computational model inspired by the structural and functional framework of the human brain. In ANNs, information is transmitted through interconnected neurons, where each input is assigned, a weight representing its contribution to the output. The weighted inputs are combined and processed through activation functions before being propagated to subsequent layers. Through iterative training, the network adjusts its weights to minimize prediction error, and the process continues until a satisfactory error level, or the maximum number of iterations is reached.

This research employed ANNs both as a baseline single-stage model and as the regression component of the proposed two-stage framework. In the single-stage setting, the ANN was trained using the entire cost range without segmentation. The network architecture was optimized through grid-based hyperparameter tuning, which systematically adjusted key model parameters to minimize validation error. Accordingly, the ANN structure was configured as follows:

Input Layer: All numerical features and one-hot encoded categorical features were normalized using Min-Max scaling and fed into the input layer.

Hidden Layers: The network architecture was selected from a predefined set of candidate configurations:

- A single hidden layer with 64 or 128 neurons, and
- A two hidden-layer architecture with 128 and 64 neurons, respectively.

Output Layer: Model training was performed using the Adam optimizer, with the initial learning rate selected from [1e-4, 1e-3, 1e-2, 5e-2] and the batch size was selected from [32, 64]. The optimization objective was the mean square error (mse).

3.2.2 Two-stage models

The two-stage framework consists of three principal components: (i) a cost-based project classification model implemented using an SVM, (ii) an ANN regressor for low-cost projects, and (iii) an ANN regressor for high-cost projects. To explicitly isolate and examine the effect of TL, this research developed two variants of the two-stage framework: one without and one with TL.

Two-stage model without TL

In the first stage, an SVM classifier was employed to partition retrofitting projects into low-cost and high-cost categories (illustrated in Figure 4). This segmentation was intended to enhance predictive performance, particularly for extreme-cost projects, which are often inadequately captured by a single regression model trained over the entire cost spectrum. The classification threshold was defined using three alternative cutoffs (i.e., mean, Q3, Q3+IQR/2), as defined in Section 3.1.

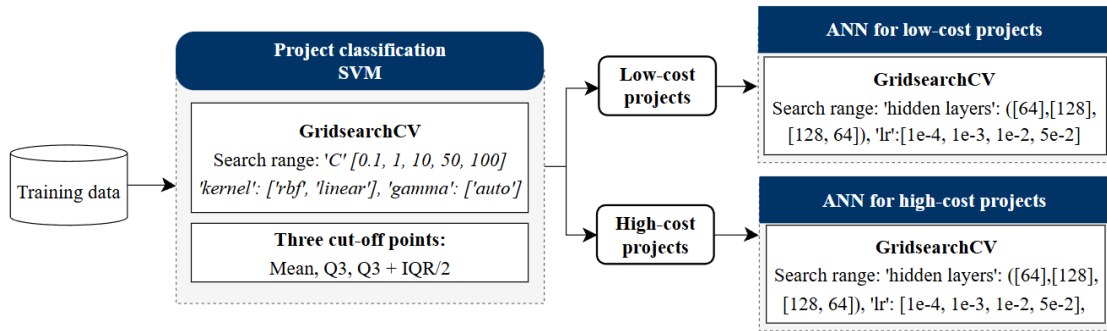


Figure 4. Components of the two-stage model without TL

In the two-stage model, the SVM aims to identify an optimal hyperplane that separates classes while maximizing the margin γ between them by solving a convex optimization problem, as described in Eqs. (3)–(6). The margin is defined as:

$$\gamma = \frac{2}{\|w\|} \quad (3)$$

x, its decision function is given by:

$$d = f(w^T x + b) \quad (4)$$

The optimization objective is:

$$\text{minimize}_{w,b} = \frac{1}{2} \|w^2\| \quad (5)$$

Subject to:

$$y_i(w^T x_i + b) \geq 1, \text{ for } i = 1, 2, \dots \quad (6)$$

For predicting retrofitting construction cost, the low-cost and high-cost ANN regressors were trained independently from scratch, without any parameter sharing or weight initialization between the two cost segments. The architectural components and data flow of this transfer-free two-stage framework are illustrated in Figure 4.

TL-enhanced two-stage model

Although the two-stage model without transfer learning (TL) can alleviate some challenges associated with wide-range cost modeling, it still has several limitations. Dividing the dataset into two subsets reduces the training samples available for each regression model, increasing the risk of overfitting, particularly for high-cost projects characterized by sparse and highly variable data. In addition, prediction errors may be amplified when the first-stage SVM misclassifies projects into incorrect cost segments. To address these issues, TL was incorporated to exploit shared information across cost segments and enhance cross-domain knowledge transfer, thereby reducing sensitivity to segmentation errors. To overcome these issues, TL was incorporated to leverage shared information across cost segments and improve prediction robustness. Specifically, the ANN trained on the low-cost segment was used as a source model to transfer learned representations to the high-cost ANN, thereby enhancing performance in the data-scarce high-cost domain. Unlike conventional ML approaches that train models independently, TL reuses knowledge from a pre-trained source model to improve prediction accuracy under limited-data conditions [30].

Formally, a domain is defined as $D = \{X, p(X)\}$, where X denotes the feature space comprising input variables (e.g., number of ground floors, floor area), and $p(X)$ represents the marginal probability distribution over X . A task associated with a domain is defined as $T = \{Y, f(\cdot)\}$, where Y is the label space (i.e., construction cost) and $f(\cdot)$ denotes the predictive function to be learned. Given a source domain D_s

with its corresponding task T_s , and a target domain D_t with task T_t , the objective of TL is to improve the target predictive function $f_T(\cdot)$ in D_t by transferring knowledge learned from D_s and T_s .

In the proposed TL-enhanced two-stage framework, the SVM classifier and the low-cost ANN (denoted as $f_{\theta_s}(x)$) were constructed identically to those in the two-stage models without TL. For high-cost prediction, a new ANN (denoted as $f_{\theta_t}(x)$) was initialized by cloning the network architecture and copying the learned parameters from the pre-trained low-cost ANN, such that $\theta_t^0 \leftarrow \theta_s$. This initialization ensured that the high-cost ANN started from a functionally meaningful solution, rather than from random weights.

To regulate the extent of knowledge transfer and domain adaptation, multiple TL strategies were examined by selectively freezing early network layers. The early layers, which capture more general feature representations, were retained as fixed, while the deeper layers were fine-tuned to learn cost-segment-specific characteristics associated with high-cost projects. Formally, the parameter set of the target model was partitioned into frozen and trainable subsets, $\theta_t = \{\theta_{\text{frozen}}, \theta_{\text{trainable}}\}$. During optimization, gradient updates were applied only to $\theta_{\text{trainable}}$, whereas θ_{frozen} remained unchanged.

Based on this framework, three TL strategies were evaluated:

- Conservative transfer: most hidden layers frozen, small learning rate.
- Balanced transfer: partial freezing with moderate learning rate.
- Aggressive transfer: all layers trainable, allow full adaptation.

3.3 Model evaluation

Model performance was evaluated using four metrics: R^2 , RMSE, and MAPE, along with visual comparisons of predicted versus observed values. These collectively offer a comprehensive view of predictive accuracy.

R^2 quantifies the proportion of variance in the target variable explained by the model. As shown in Eq. (7), y_i is the observed value, $\hat{y}_i$ is the predicted value, and $\bar{y}$ is the mean of the observed values. R^2 ranges from 0 to 1, with higher values indicating better model performance, though it should be interpreted alongside other metrics due to its sensitivity to variance.

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2} \quad (7)$$

RMSE, defined in Eq. (8), quantifies the average magnitude of prediction errors and assigns greater penalties to large errors.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (8)$$

MAPE evaluates the average percentage error, calculated in Eq. (9).

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (9)$$

3.4 Model interpretation with SHAP

Due to the complexity of the proposed TL-enhanced two-stage framework, post hoc interpretability was required to explain the prediction behavior of the low- and high-cost models. Accordingly, SHAP, based on Shapley values from cooperative game theory, was employed to quantify the contribution of each input feature to the model output [48]. By evaluating each feature's marginal contribution across different feature combinations, SHAP provides consistent and theoretically grounded model interpretations.

The SHAP explanation model is expressed in Eq. (10):

$$q(s) = \theta_0 + \sum_{j=1}^M \theta_j s_j, \quad (10)$$

where $q(s)$ denotes the model, M is the total number of input features, θ_j represents the SHAP value associated with the j -th feature, and θ_0 is a model constant.

4. Results and discussion

4.1 Model performance

The average performance of all models in test set across 10 different random seeds is summarized in Table 4. For the single-model group, RMSE values range from 28,750 to 45,598 (USD), with the models explaining approximately 80.77% to 91.73% of the variance in the test set. Although the overall predictive performance of these models is generally acceptable, their segment-wise performance deteriorates substantially for high-cost projects. In particular, when the project cost exceeds Q3, the RMSE increases to 75,394, and further escalates to 90,029 when the cost exceeds $Q3 + IQR/2$, as observed for the Monotonic GBM model. This degradation can be attributed to the presence of extreme cost values that are retained in the dataset rather than removed as outliers, which disproportionately affects model accuracy in high-cost segments. Notably, although Monotonic GBM incorporates domain knowledge-based monotonic constraints, these constraints do not result in superior performance compared with other single models.

Among the single models, XGBoost, CatBoost, and ANN consistently outperform Monotonic GBM across all evaluation metrics. The ANN model exhibits the strongest performance, achieving an average R^2 of 0.9173 with a noticeably smaller standard deviation across random seeds compared with other single models. Its MAPE is reduced to 20.41 ± 6.92 (%), and both the overall RMSE and segment-wise RMSE values are substantially lower than those of the boosting-based models. These results indicate that ANN-based approaches are more robust to wide-range cost distributions and extreme values than tree-based boosting algorithms. The superiority of the ANN model is further illustrated in Figure 5. While CatBoost, XGBoost, and Monotonic GBM exhibit lower average performance and higher variability across random seeds, the ANN model demonstrates consistently higher accuracy with reduced performance dispersion, highlighting its stability and suitability for retrofitting cost prediction under heterogeneous cost conditions.

Table 4. Performance of single and two-stage models

Model	Cut off	Transfer strategy	R^2	MAPE (%)	RMSE	RMSE > Q3	RMSE > Q3+IQR/2
CatBoost			0.8789 ± 0.08	15.38 ± 1.49	$35,718 \pm 13,735$	62,782	75,493
XGBoost			0.8675 ± 0.04	16.00 ± 2.64	$37,888 \pm 9,780$	66,928	80,457
Monotonic GBM			0.8077 ± 0.06	26.75 ± 3.19	$45,598 \pm 11,897$	75,394	90,029
ANN			0.9173 ± 0.03	20.41 ± 6.92	$28,750 \pm 4,161$	41,163	47,410
Two-stage no transfer	Mean		0.8617 ± 0.09	18.99 ± 7.74	$35,488 \pm 9,933$	44,755	53,035
	Q3		0.8740 ± 0.09	18.88 ± 7.78	$33,734 \pm 10,07$	40,609	47,806
	Q3+IQR/2		0.8639 ± 0.09	18.49 ± 7.45	$35,575 \pm 9,926$	46,813	56,142
Two-stage transfer	Mean	Conservative	0.9568 ± 0.01	10.79 ± 1.03	$21,360 \pm 3,084$	34,514	40,195
		Balanced	0.9566 ± 0.01	10.95 ± 1.12	$21,377 \pm 2,840$	34,337	39,797
		Aggressive	0.9519 ± 0.02	11.25 ± 1.35	$22,068 \pm 2,810$	34,998	41,994
	Q3	Conservative	0.9506 ± 0.01	12.37 ± 1.86	$22,788 \pm 3,310$	38,032	44,628
		Balanced	0.9508 ± 0.01	12.74 ± 2.03	$22,584 \pm 2,495$	37,309	43,041
		Aggressive	0.9370 ± 0.03	14.26 ± 3.23	$24,912 \pm 3,612$	39,153	46,276
	Q3+IQR/2	Conservative	0.9520 ± 0.01	12.29 ± 2.01	$22,404 \pm 2,679$	37,396	44,173
		Balanced	0.9546 ± 0.01	12.46 ± 2.06	$21,722 \pm 2,042$	35,786	42,010
		Aggressive	0.9334 ± 0.04	13.48 ± 3.18	$25,460 \pm 5,809$	42,185	50,714

For the two-stage model group, the models without TL achieve R^2 values ranging from 0.86 to 0.87, which is comparable to the performance of CatBoost and XGBoost. Interestingly, when focusing on high-cost project segments ($\geq Q3$ and $\geq Q3 + IQR/2$), the two-stage models without TL exhibit substantially lower RMSE, as illustrated in Figure 5(c) and (d). This improvement can be attributed to cost-based segmentation, which mitigates the influence of extreme values by separating projects into distinct cost ranges, thereby enabling more stable learning for high-cost cases. For two-stage models without TL, varying the cutoffs used for cost segmentation does not result in pronounced performance differences (Table 4). Nevertheless, despite these improvements in high-cost segments, splitting the dataset significantly reduces the number of training samples available for each model, which in turn limits the learning capacity of the regressors. As a consequence, the overall predictive performance of the two-stage models without TL remains inferior to that of the single ANN model, as evidenced in Figure 5 (a) and (b).

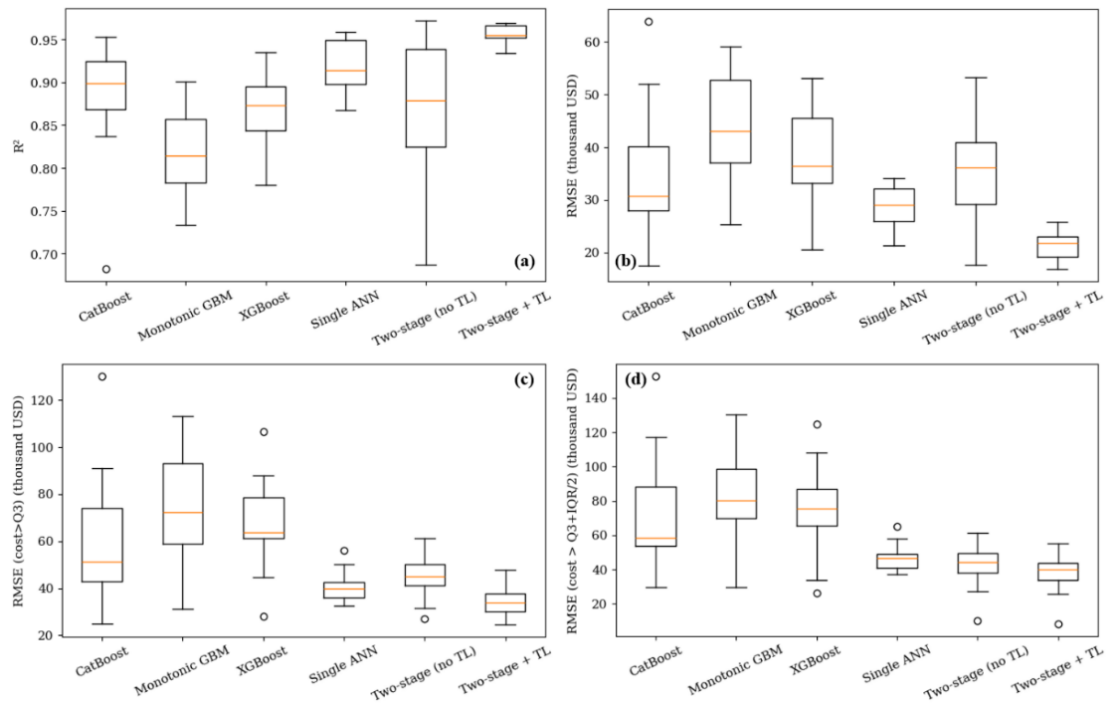


Figure 5. Comparison of the predictive performance of different models.

Note: Due to the large number of cutoffs and TL strategies, the two-stage models are represented by (1) a two-stage framework without TL using the mean cost as the cutoff, and (2) a TL-enhanced two-stage framework using the mean cost as the cutoff, to facilitate clearer comparison across model groups.

Compared with the single models and the two-stage models without TL, the proposed TL-enhanced two-stage models consistently achieve the best predictive performance, regardless of the selected TL strategy or cost cutoff. Among the evaluated two-stage TL strategies, the model combining the mean-based cutoff with the balanced TL strategy yields the highest average performance. When compared with the best-performing baseline model (i.e.,

single ANN), the TL-enhanced two-stage framework reduces MAPE from 20.41% to 10.95% and overall RMSE from USD 28,750 to USD 21,377, corresponding to reductions of 46% and 25%, respectively. For high-cost segments, RMSE decreases from USD 41,163 to USD 34,337 for projects with cost $\geq Q3$ and from USD 47,410 to USD 39,797 for projects with cost $\geq Q3 + IQR/2$, corresponding to reductions of 17% and 16%, respectively. These results demonstrate the effectiveness of transfer learning in improving predictive reliability, particularly for high-cost retrofit projects.

Figure 6 compares the residual distributions of the TL-enhanced two-stage model and the single ANN. The residuals of the two-stage transferred ANN are largely concentrated within the range of $-100,000$ to $100,000$ (USD), whereas the single ANN exhibits substantially larger errors, with residuals extending to approximately $-200,000$ to $200,000$. Residuals are defined as the difference between the observed values (y_{test}) and the predicted values (y_{pred}). The observed pattern indicates that the single ANN tends to overestimate costs for low-cost projects and underestimate costs for high-cost projects, a systematic bias that is effectively mitigated by the proposed TL-enhanced two-stage framework.

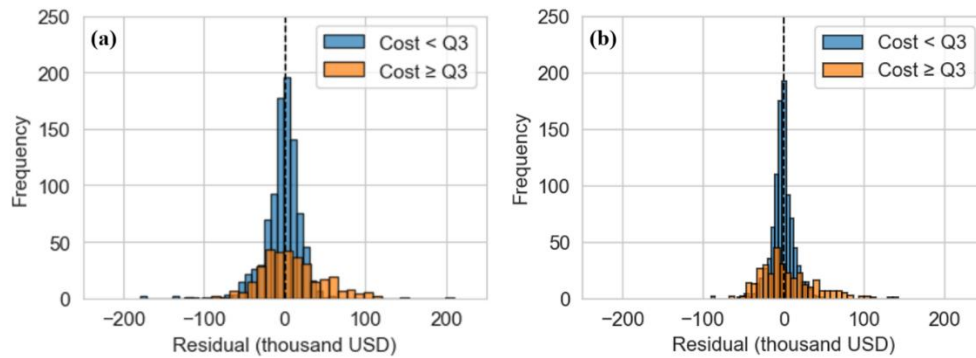


Figure 6. Residual of (a) single ANN and (b) two-stage transferred model (using balanced TL and mean-based cutoff) per cost segment

The comparison in Figure 7 indicates that the two-stage transferred model achieves superior performance compared to the single ANN model. The two-stage model more accurately follows the trend of the actual data, capturing both peaks and local fluctuations. Meanwhile, the single ANN model frequently underestimates extreme values, reflecting weaker generalization capability.

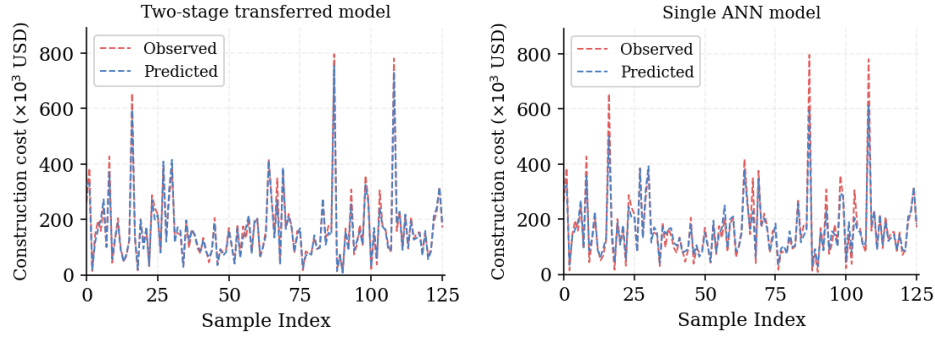


Figure 7. Prediction in a single random seed of single ANN and two-stage transferred model (using balanced TL and mean-based cutoff) per cost segment

Overall, the TL-enhanced two-stage models consistently outperform the baseline models. However, their average predictive performance varies depending on the selected TL strategy and cost cutoff, indicating that the effects of these two factors warrant further investigation to establish a generalizable and robust framework.

Regarding the impact of cost cutoffs, a Wilcoxon signed-rank test was conducted to examine whether performance differences among models using different cutoff values are statistically significant. The results, summarized in Table 5, show that models employing the mean-based cutoff exhibit statistically significant performance differences compared with those using Q3 and Q3 + IQR/2 cutoffs, with all evaluation metrics yielding p-values below 0.05. In contrast, no statistically significant differences are observed between the models using Q3 and Q3 + IQR/2 cutoffs. Combined with the performance results presented in Table 4, these findings indicate that the mean-based cutoff provides the most favorable and stable performance among the three cutoffs evaluated.

Table 5. p-values from the Wilcoxon signed-rank test for two-stage transferred models with different cutoff thresholds

Metric	Mean and Q3	Mean and Q3+IQR/2	Q3 and Q3+IQR/2
R^2	1.19E-06	5.38E-03	1.46E-01
RMSE	1.19E-06	7.11E-03	1.40E-01
MAPE	1.86E-09	3.15E-07	2.05E-01
RMSE (cost > Q3)	3.79E-06	1.60E-05	6.26E-01
RMSE (cost > Q3+IQR/2)	1.19E-06	3.45E-05	7.77E-01

The results in Table 4 indicate that the selection of cost cutoff primarily affects model performance in the high-cost segments, as the RMSE values for projects with cost $\geq$ Q3 and cost $\geq$ Q3 + IQR/2 exhibit more pronounced differences than the RMSE computed over the entire dataset. To further investigate the influence of cutoff selection, its impact on the SVM-based cost range classification was analyzed.

Figure 8 presents the classification performance of the SVM models under different cutoff strategies. The SVM using the Q3 + IQR/2 cutoff achieves the highest precision, occasionally approaching a value of 1.0. However, this comes at the expense of low recall and high variability across runs, resulting in an unstable F1-score. In contrast, the Q3 cutoff yields the lowest average precision, recall, and F1-score among the three cutoff strategies. The mean-based cutoff provides a more balanced trade-off between precision and recall. Although its precision is lower than that of the Q3 + IQR/2 cutoff, it achieves the highest average F1-score with the lowest variability, indicating more stable and reliable classification performance. These results suggest that the mean cutoff represents the safest and most robust choice among the evaluated cutoff strategies for cost range classification within the proposed two-stage framework.

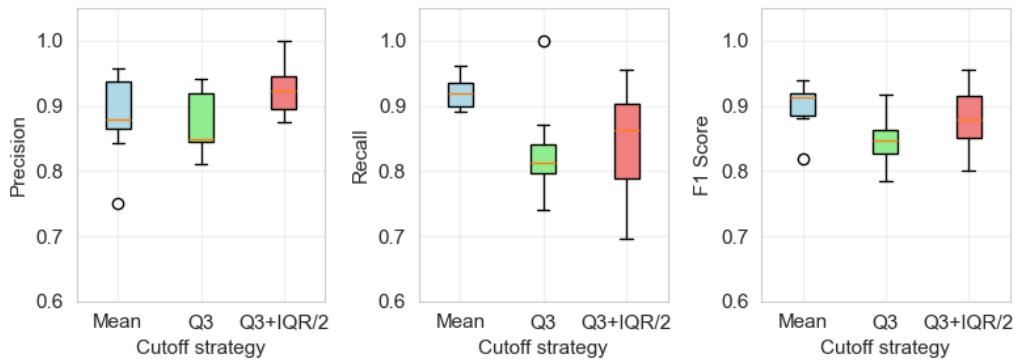


Figure 8. Classification performance of SVM using different cutoffs

The observed differences among the SVM classifiers using different cutoffs can be attributed primarily to class distribution effects. As illustrated in Figure 9, the mean-based cutoff provides a relatively balanced distribution between low- and high-cost projects, whereas the Q3 and Q3 + IQR/2 cutoffs create increasingly imbalanced ratios of approximately 75:25 and 84:16, respectively. This imbalance causes high-cost projects to be more frequently misclassified as low-cost, leading to error propagation in the second-stage regressors. Consequently, models using highly imbalanced cutoffs exhibit higher RMSE in the high-cost segment. In contrast, the mean-based cutoff reduces misclassification risk and provides more stable prediction performance for high-cost projects.

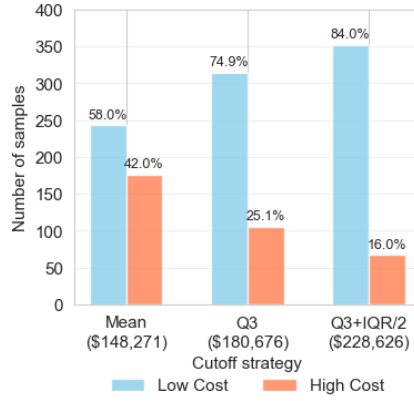


Figure 9. Class distributions of low-cost and high-cost projects under different cutoffs

Regarding the influence of TL strategies, the models were grouped according to the adopted TL strategy and evaluated using the Wilcoxon signed-rank test to examine whether different strategies lead to statistically significant differences in predictive performance. The results reported in Table 6 indicate that the model employing the balanced TL strategy achieves significantly improved prediction performance in the high-cost segments, with p-values below 0.05 for RMSE when $\text{cost} \geq Q3$ and $\text{cost} \geq Q3 + IQR/2$, compared with the other strategies. In contrast, the conservative and aggressive strategies do not show statistically significant improvements. The conservative strategy relies too heavily on knowledge transferred from low-cost data, limiting adaptation to high-cost project characteristics, whereas the aggressive strategy retains insufficient pre-trained knowledge, reducing generalization capability and resulting in the highest average RMSE in the high-cost region.

Table 6. p-values from the Wilcoxon signed-rank test for two-stage transferred models with different TL strategies

Metric	Conservative vs Balanced	Conservative vs Aggressive	Balanced vs Aggressive
R^2	1.58E-01	2.99E-03	1.23E-03
RMSE	1.46E-01	1.13E-02	2.196E-03
MAPE	2.55E-07	7.98E-04	6.19 E-03
RMSE ($\text{cost} > Q3$)	2.93E-02	1.52E-01	4.27 E-02
RMSE ($\text{cost} > Q3+IQR/2$)	3.22E-03	1.14E-01	2.34 E-02

4.2 Model interpretation

The TL-enhanced two-stage model using balanced TL with a mean-based cutoff was further analyzed using SHAP. First, the behavior of the SVM classifier was examined, as illustrated in Figure 10. The results indicate that insulation-related features, including N6 (external wall insulation), N8 (roof insulation work quantity), and N14 (high-efficiency HVAC system installation cost), exert the strongest influence on the classification outcome. Higher values of these features substantially increase the likelihood that a project is classified into the high-cost

category. In addition, building type plays a significant role in the classifier’s decision-making process: nursery buildings tend to be classified as high-cost, whereas healthcare centers are more frequently assigned to the low-cost category. In contrast, features related to construction area and time, such as N9 (floor work quantity), N3 (gross floor area), N19 (solar radiation control device), and N1 (year of completion), exhibit relatively weaker contributions to the SVM’s classification decisions.

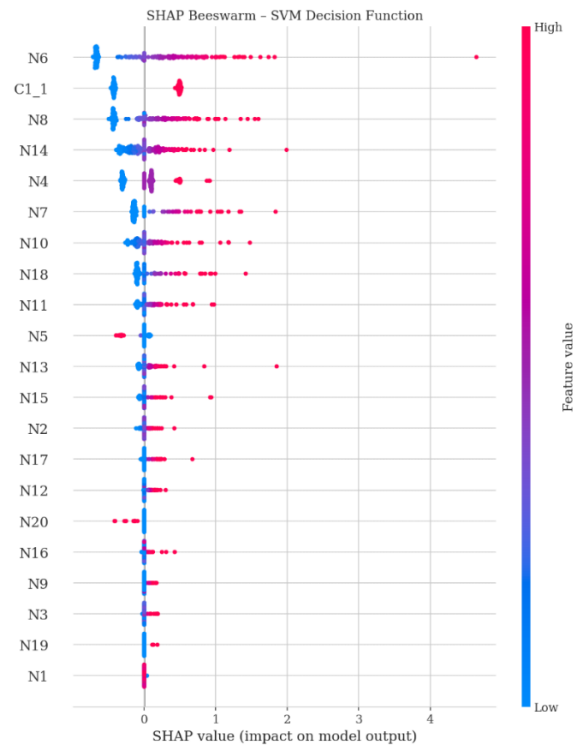


Figure 10. SHAP values of the SVM classifier

Figure 11 illustrates the SHAP summary plots of the ANN regressors for low-cost and high-cost projects, revealing both shared patterns and segment-specific differences in feature influence. In the low-cost ANN, the model is primarily driven by envelope- and insulation-related variables, with N6 (external wall insulation), N14 (high-efficiency HVAC system installation cost), N10 (window replacement cost), and N8 (roof insulation work quantity) occupying the top positions in the importance ranking. These variables exhibit wide SHAP distributions and clear monotonic trends, indicating their dominant and stable contribution to cost prediction in lower-budget projects. When transitioning to the high-cost ANN, although N6 remains the most influential feature, notable reordering among the top-ranked variables is observed. Specifically, N10 and N14 exchange relative positions, while N7 (internal wall insulation) and N18 (renewable energy system installation quantity) move upward in the ranking. This shift indicates that high-cost projects place greater emphasis on integrated insulation strategies and energy-generation systems rather than solely on basic envelope upgrades. Moreover, energy-related variables such as N18

(solar PV installation quantity), N15 (high-efficiency boiler installation cost), and N16 (LED lighting replacement quantity) show increased SHAP dispersion and higher positive contributions in the high-cost segment compared to the low-cost model. Although their absolute SHAP magnitudes remain smaller than those of major insulation variables, their elevated ranking reflects a change in the dominant cost formation mechanism, where advanced energy retrofit measures play a more critical role. These differences in feature importance provide strong support for the proposed two-stage framework, demonstrating that separate ANN regressors can better capture distinct cost formation mechanisms across low- and high-cost project segments.

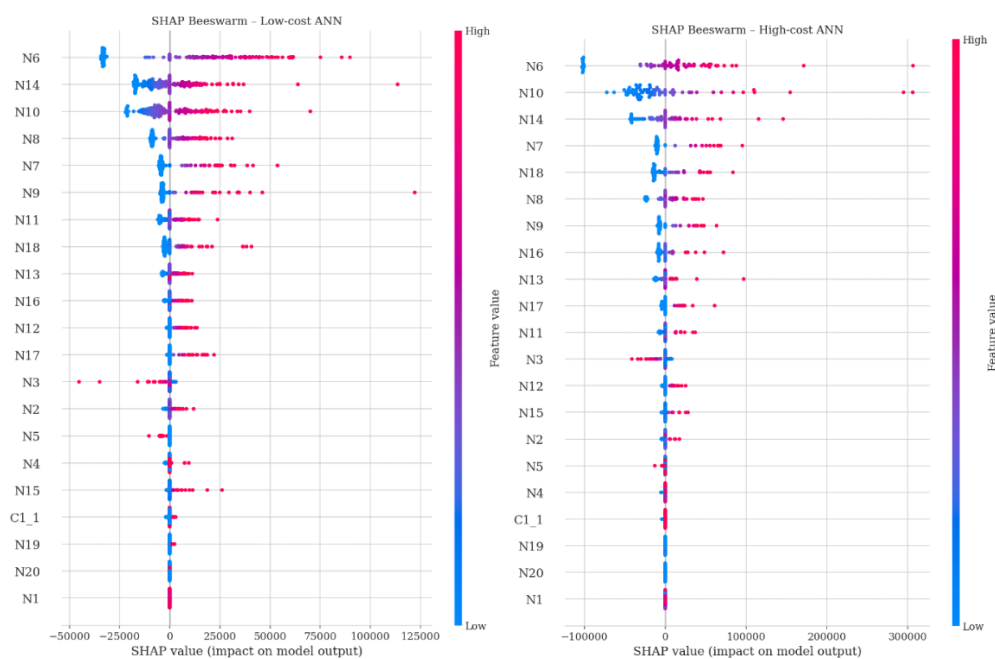


Figure 11. SHAP values of the ANN regressor for low-cost projects and high-cost projects

Overall, the SHAP interpretation indicates that the ANN models prioritize input variables in a manner consistent with the practical sequence and technical importance of retrofitting activities rather than relying on abstract building descriptors. Envelope- and insulation-related measures, such as external wall insulation, window replacement, and roof insulation, consistently emerge as the most influential factors, reflecting their role as early-stage interventions that establish the thermal and structural basis for subsequent upgrades and largely determine the overall retrofitting cost. As the retrofitting process progresses, energy system-related measures (e.g., including high-efficiency HVAC systems, boilers, and renewable energy installations) gain influence, particularly in high-cost projects, aligning with real-world construction workflows in which mechanical and electrical upgrades are implemented after envelope performance has been secured. In contrast, conventional descriptors such as

construction year, total floor area, and number of stories exhibit limited contributions across both cost segments, suggesting that retrofitting costs are governed more by the type, sequence, and technical complexity of retrofit measures than by building scale alone.

5. Implications

The results demonstrate that the proposed TL-enhanced two-stage model provides reliable and accurate predictions, achieving an average R^2 of 0.9566 on the test set and substantially reducing RMSE for high-cost project segments compared with conventional models. Beyond predictive performance, the framework offers several practical implications for data-driven decision-support in construction management.

The two-stage architecture is particularly effective for early-stage retrofitting cost estimation, where detailed cost information is often unavailable (Figure 12). By separating low- and high-cost projects, the framework reduces the risk of cost underestimation in high-value projects and supports more reliable planning decisions. Correlation and SHAP analyses further show that the “year of completion” has limited influence on predictions, indicating temporal robustness, while cost-sensitive retrofit variables such as window, door, and HVAC replacement better capture changes in material, labor, and construction practices over time.

Although validated on nursery and healthcare buildings, the framework is scalable to other building types and regional contexts through retraining with locally relevant datasets and recalibration of cost-related variables. SHAP-based interpretability also provides a mechanism to verify whether learned cost patterns remain consistent across different applications. More broadly, this research presents a general AI-driven modeling paradigm for construction analytics problems with highly dispersed target distributions, where removing outliers may result in significant information loss. By integrating SVM-based segmentation, ANN regression, and transfer learning, the framework effectively captures heterogeneous cost patterns, particularly in underrepresented high-cost segments. However, successful application requires careful adaptation of segmentation thresholds and TL strategies to local data characteristics, along with rigorous evaluation of the classification stage and post-hoc interpretability analysis to ensure transparent and reliable deployment in construction decision-support systems (Figure 12).

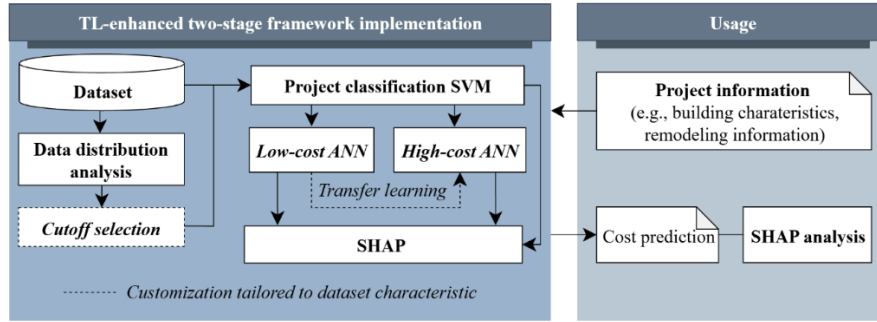


Figure 12. Implementation workflow of the TL-enhanced two-stage framework

6. Conclusions

This research developed a segment-aware transfer learning framework for reliable early-stage building retrofit cost estimation under skewed cost distributions. Using a real-world dataset with a highly right-skewed cost distribution, projects were first classified into low- and high-cost groups using SVM, followed by segment-specific ANN regressors. Transfer learning was applied to transfer knowledge from the data-rich segment to the data-scarce segment. Using a real-world dataset with a highly right-skewed cost distribution, projects were first classified into low- and high-cost groups using SVM, followed by segment-specific ANN regressors. TL was applied to transfer knowledge from the data-rich segment to the data-scarce segment. The proposed framework, including different cost cutoffs and TL strategies, was systematically compared with conventional single-stage models such as CatBoost, XGBoost, Monotonic GBM, and ANN. Based on the experimental results, the following conclusions can be drawn:

- All TL-enhanced two-stage models consistently outperform the single-stage models, achieving reductions of 46% in MAPE and 25% in RMSE on the overall test set.
- Although two-stage models without TL are specialized for each cost segment, their overall performance is inferior to that of the single ANN model due to the limited number of training samples available for each regressor.
- Cost cutoff selection should prioritize class balance to ensure reliable SVM classification performance, rather than maximizing data availability for the pre-trained model. Under data-scarce conditions, as in this research, the mean-based cutoff represents the safest and most robust choice.

- The balanced TL strategy is the most effective, yielding the most pronounced improvement in RMSE for high-cost projects. In contrast, the aggressive TL strategy is potentially risky, particularly when combined with cutoffs that induce severe class imbalance.
- SHAP analysis reveals that insulation- and energy-related variables dominate the model's predictive behavior, whereas variables related to construction year, floor area, or the number of stories play a comparatively minor role.

In the context of limited data availability and highly heterogeneous cost distributions in construction projects, the proposed framework provides a data-driven decision-support tool for early-stage retrofitting cost analysis. By retaining extremely high-cost projects during model training rather than excluding them, the framework preserves critical project information and improves the reliability of automated cost estimation, thereby reducing the risk of cost underestimation and subsequent budget overruns compared with conventional prediction approaches.

Despite these contributions, the framework has only been validated on nursery and healthcare retrofitting projects. Expanding the dataset to additional building types would improve the robustness and generalizability of the proposed system. Furthermore, since the current dataset mainly contains numerical and categorical variables, future research may incorporate unstructured textual project information using natural language processing and large language models to further enhance automated construction cost analytics and decision-support capabilities.

Acknowledgements

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) (RS-2024-00344868).

References

- [1] C. Aparicio-Fernández, M. E. Torner, M. Cañada-Soriano, and J.-L. Vivancos, "Analysis of the energy performance strategies in a historical building used as a music school," *Developments in the Built Environment*, vol. 15, 2023, doi: 10.1016/j.dibe.2023.100195.
- [2] S. Park, S. Park, H. Jang, Y. Ahn, and N. Kwon, "Computing green remodeling construction cost for public buildings based on genetic algorithm and case-based reasoning," *Developments in the Built Environment*, vol. 22, p. 100655, 2025.
- [3] L. Pérez-Lombard, J. Ortiz, and C. Pout, "A review on buildings energy consumption information," *Energy and Buildings*, vol. 40, no. 3, pp. 394-398, 2008, doi: 10.1016/j.enbuild.2007.03.007.

- [4] C. Fan, F. Xiao, Z. Li, and J. Wang, "Unsupervised data analytics in mining big building operational data for energy efficiency enhancement: A review," *Energy and Buildings*, vol. 159, pp. 296-308, 2018, doi: 10.1016/j.enbuild.2017.11.008.
- [5] M. Aksoezen, M. Daniel, U. Hassler, and N. Kohler, "Building age as an indicator for energy consumption," *Energy and Buildings*, vol. 87, pp. 74-86, 2015, doi: 10.1016/j.enbuild.2014.10.074.
- [6] Z. Ma, P. Cooper, D. Daly, and L. Ledo, "Existing building retrofits: Methodology and state-of-the-art," *Energy and buildings*, vol. 55, pp. 889-902, 2012.
- [7] P. Xu, E. H. W. Chan, H. J. Visscher, X. Zhang, and Z. Wu, "Sustainable building energy efficiency retrofit for hotel buildings using EPC mechanism in China: analytic Network Process (ANP) approach," *Journal of Cleaner Production*, vol. 107, pp. 378-388, 2015, doi: 10.1016/j.jclepro.2014.12.101.
- [8] Y. Fan and X. Xia, "Energy-efficiency building retrofit planning for green building compliance," *Building and Environment*, vol. 136, pp. 312-321, 2018, doi: 10.1016/j.buildenv.2018.03.044.
- [9] N. Kwon, Y. Ahn, B.-S. Son, and H. Moon, "Developing a machine learning-based building repair time estimation model considering weight assigning methods," *Journal of Building Engineering*, vol. 43, 2021, doi: 10.1016/j.jobe.2021.102627.
- [10] D. W. M. Chan, "Sustainable building maintenance for safer and healthier cities: Effective strategies for implementing the Mandatory Building Inspection Scheme (MBIS) in Hong Kong," *Journal of Building Engineering*, vol. 24, 2019, doi: 10.1016/j.jobe.2019.100737.
- [11] D. E. Ighravwe and S. A. Oke, "A multi-criteria decision-making framework for selecting a suitable maintenance strategy for public buildings using sustainability criteria," *Journal of Building Engineering*, vol. 24, 2019, doi: 10.1016/j.jobe.2019.100753.
- [12] N. Kwon, K. Song, Y. Ahn, M. Park, and Y. Jang, "Maintenance cost prediction for aging residential buildings based on case-based reasoning and genetic algorithm," *Journal of Building Engineering*, vol. 28, p. 101006, 2020.
- [13] D. Monfet, M. Corsi, D. Choinière, and E. Arkhipova, "Development of an energy prediction tool for commercial buildings using case-based reasoning," *Energy and Buildings*, vol. 81, pp. 152-160, 2014.
- [14] R. Ruparathna, K. Hewage, and R. Sadiq, "Economic evaluation of building energy retrofits: A fuzzy based approach," *Energy and Buildings*, vol. 139, pp. 395-406, 2017.

- [15] M. A. Yamusa, H. S. i. Lawal, R. S. Abdulrahman, A. S. Salisu, A. Saka, M. Abubakar, and M. Abdullahi, "A machine-learning model for estimating construction renovation costs," *International Journal of Building Pathology and Adaptation*, 2025.
- [16] J. Kim and H. Cha, "Expediting the Cost Estimation Process for Aged-Housing Renovation Projects Using a Probabilistic Deep Learning Approach, Sustainability, 2022," ed: DOI.
- [17] S. Mulero-Palencia, S. Álvarez-Díaz, and M. Andrés-Chicote, "Machine learning for the improvement of deep renovation building projects using as-built BIM models," *Sustainability*, vol. 13, no. 12, p. 6576, 2021.
- [18] V. E. Papadimitriou and G. N. Aretoulis, "A final cost estimating model for building renovation projects," *Buildings*, vol. 14, no. 4, p. 1072, 2024.
- [19] B. J. Gardner, D. D. Gransberg, and J. A. Rueda, "Stochastic conceptual cost estimating of highway projects to communicate uncertainty using bootstrap sampling," *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, vol. 3, no. 3, p. 05016002, 2017.
- [20] M. Juszczak, T. Hanák, M. Výskala, H. Pacyno, and M. Siejda, "Early fast cost estimates of sewerage projects construction costs based on ensembles of neural networks," *Applied Sciences*, vol. 13, no. 23, p. 12744, 2023.
- [21] S. Mahmoud, M. F. Khamidi, A. Idrus, and O. A.-L. Ashola, "Development of maintenance cost prediction model for heritage buildings," *Jurnal Teknologi (Sciences & Engineering)*, vol. 74, no. 2, 2015.
- [22] C.-S. Li and S.-J. Guo, "Development of a cost predicting model for maintenance of university buildings," in *Proceedings of the 2011 2nd International Congress on Computer Applications and Computational Science: Volume 1*, 2012: Springer, pp. 215-221.
- [23] A. G. Mohamed, J. E. Ghaly, and M. Marzouk, "Revolutionizing semantic integration of maintenance cost prediction for building systems using artificial neural networks," *Journal of Building Engineering*, vol. 96, p. 110416, 2024.
- [24] J.-M. Kim, S.-G. Yum, M. D. Adhikari, and J. Bae, "A LSTM algorithm-driven deep learning approach to estimating repair and maintenance costs of apartment buildings," *Engineering, Construction and Architectural Management*, vol. 31, no. 13, pp. 369-389, 2024.
- [25] Z. Wu, M. Liu, G. Ma, and S. Jiang, "A hybrid forecasting model to improve cost prediction accuracy in green building projects with machine learning," *Engineering, Construction and Architectural Management*, 2025.

- [26] Z. Wu, M. Liu, and G. Ma, "A machine learning-based two-stage integrated framework for cost reasonableness prediction of green building projects," *Journal of Building Engineering*, vol. 100, p. 111733, 2025.
- [27] K. S. Al-Gahtani, A. M. Alsugair, N. M. Alsanabani, A. A. Alabduljabbar, and A. S. Almohsen, "ANN prediction model of final construction cost at an early stage," *Journal of Asian Architecture and Building Engineering*, vol. 24, no. 2, pp. 775-799, 2025.
- [28] H. A. T. Nguyen, D. H. Pham, Y. Ahn, B. L. Oo, and B. T. H. Lim, "Machine learning and sustainable geopolymer materials: A systematic review," *Materials Today Sustainability*, p. 101095, 2025.
- [29] D. R. Das Adhikary, R. Sahu, and S. Pragyna Panda, "Prediction of used car prices using machine learning," in *Biologically Inspired Techniques in Many Criteria Decision Making: Proceedings of BITMDM 2021*: Springer, 2022, pp. 131-140.
- [30] H. A. T. Nguyen, D. H. Pham, A. T. Le, Y. Ahn, B. L. Oo, and B. T. H. Lim, "Transfer learning framework for modelling the compressive strength of ultra-high performance geopolymer concrete," *Construction and Building Materials*, vol. 459, p. 139746, 2025.
- [31] C. Hennebold, K. Klöpfer, P. Lettenbauer, and M. Huber, "Machine learning based cost prediction for product development in mechanical engineering," *Procedia CIRP*, vol. 107, pp. 264-269, 2022.
- [32] V. E. Papadimitriou and G. N. Aretoulis, "An Innovative Approach Regarding Efficient and Expedited Early Building Renovation Cost Estimation Utilizing ANNs and the TOPSIS Methodology," *Algorithms*, vol. 18, no. 11, p. 696, 2025.
- [33] C. Deb, Z. Dai, and A. Schlueter, "A machine learning-based framework for cost-optimal building retrofit," *Applied energy*, vol. 294, p. 116990, 2021.
- [34] H. Nasrazadani, M. Mahsuli, H. Talebiyan, and H. Kashani, "Probabilistic modeling framework for prediction of seismic retrofit cost of buildings," *Journal of Construction Engineering and Management*, vol. 143, no. 8, p. 04017055, 2017.
- [35] Y. Gao, Y. Ruan, C. Fang, and S. Yin, "Deep learning and transfer learning models of energy consumption forecasting for a building with poor information data," *Energy and Buildings*, vol. 223, p. 110156, 2020.
- [36] E. Ford, K. Maneparambil, A. Kumar, G. Sant, and N. Neithalath, "Transfer (machine) learning approaches coupled with target data augmentation to predict the mechanical properties of concrete," *Machine Learning with Applications*, vol. 8, p. 100271, 2022.

- [37] H. Pak and S. G. Paal, "Evaluation of transfer learning models for predicting the lateral strength of reinforced concrete columns," *Engineering Structures*, vol. 266, p. 114579, 2022.
- [38] H. Ji, Y. Tian, C. Fu, and H. Ye, "Transfer learning enables prediction of steel corrosion in concrete under natural environments," *Cement and Concrete Composites*, vol. 148, p. 105488, 2024.
- [39] Q. Yang, W. Shi, J. Chen, and W. Lin, "Deep convolution neural network-based transfer learning method for civil infrastructure crack detection," *Automation in Construction*, vol. 116, p. 103199, 2020.
- [40] Y. Guo, H. Niu, and S. Li, "Safety monitoring in construction site based on unmanned aerial vehicle platform with computer vision using transfer learning techniques," in *Asia-Pacific Workshop on Structural Health Monitoring*, 2018, vol. 7.
- [41] H. Luo, J. Chen, P. E. Love, and W. Fang, "Explainable transfer learning for modeling and assessing risks in tunnel construction," *IEEE Transactions on Engineering Management*, vol. 71, pp. 8339-8355, 2024.
- [42] H. A. T. Nguyen, D. H. Pham, B. Kim, Y. Ahn, and N. Kwon, "Developing an automated framework for eco-label information categorization using web crawling and Natural Language Processing techniques," *Expert Systems with Applications*, p. 127688, 2025.
- [43] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," *Advances in neural information processing systems*, vol. 31, 2018.
- [44] C. Bentéjac, A. Csörgő, and G. Martínez-Muñoz, "A comparative analysis of gradient boosting algorithms," *Artificial Intelligence Review*, vol. 54, pp. 1937-1967, 2021.
- [45] S.-B. Lee, Y.-J. Kim, S. Hwang, H. Son, S. K. Lee, K.-I. Park, and Y.-G. Kim, "Predicting Parkinson's disease using gradient boosting decision tree models with electroencephalography signals," *Parkinsonism & Related Disorders*, vol. 95, pp. 77-85, 2022.
- [46] S. Demir and E. K. Sahin, "Predicting occurrence of liquefaction-induced lateral spreading using gradient boosting algorithms integrated with particle swarm optimization: PSO-XGBoost, PSO-LightGBM, and PSO-CatBoost," *Acta Geotechnica*, vol. 18, no. 6, pp. 3403-3419, 2023.
- [47] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 2016, pp. 785-794.
- [48] A. E. Roth, "Introduction to the Shapley value," *The Shapley value*, vol. 1, 1988.